The Hénon–Heiles Problem

Nonlinear Dynamics, Symplectic Integration, and the Edge of Chaos

Companion Guide to henon_heiles.ipynb

Contents

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1. The Scientific Question

In the early 1960s, astronomers studying the motion of stars within galaxies faced a nagging theoretical puzzle. Classical mechanics teaches that a dynamical system possesses as many independent conserved quantities — *integrals of motion* — as it has degrees of freedom. For a star moving in an axisymmetric galactic potential, two such integrals are obvious: total energy, and angular momentum about the symmetry axis. Astronomers had observed, however, that the distribution of stellar velocities in the solar neighborhood was far more constrained than these two integrals alone could explain. Stars moved as though something was holding them on tighter orbital leashes than energy and angular momentum permitted. This suggested the existence of a mysterious *third integral* of motion — some hidden conserved quantity keeping the orbits better-behaved than they had any right to be.

The problem was that nobody could write this third integral down in closed form. It did not appear to have a simple algebraic expression. Michel Hénon and Carl Heiles decided to attack the question numerically. Rather than study a full three-dimensional galactic potential — computationally intractable on the hardware of 1964 — they stripped the problem to its mathematical skeleton: the simplest possible two-dimensional nonlinear oscillator that could model the essential physics. Their goal was to determine whether the third integral actually existed in general, or whether it was only an approximate feature of orbits at low energy that dissolved into chaos at higher energies.

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2. The Two Scientists

Michel Hénon (1931–2013)

Michel Hénon was a French astronomer and mathematician at the Observatoire de Nice. He is one of the great figures of twentieth-century dynamical systems theory, and his career spanned an extraordinary range: from the statistical mechanics of star clusters and the gravitational three-body problem, to fluid turbulence, to pure combinatorics. He is perhaps best known outside astronomy for the *Hénon map* — a two-dimensional discrete dynamical system that became one of the canonical examples of a strange attractor, a fractal object generated by deterministic chaos. Hénon was unusual among physicists of his era in treating numerical computation as a tool for mathematical *discovery* rather than merely a calculator for confirming known results. The 1964 paper with Heiles was an early and exemplary instance of this philosophy: use the computer not to verify a theory, but to find out what actually happens.

Carl Heiles (1939–)

Carl Heiles is an American radio astronomer, long associated with the University of California, Berkeley. His primary scientific contributions have been in interstellar medium research — mapping the structure of neutral hydrogen in the Milky Way and characterizing the diffuse gas between the stars. The Hénon–Heiles paper was, in a sense, a detour from his main research program, but it became the work for which he is most widely recognized outside radio astronomy. Heiles brought to the collaboration both the astronomical motivation (the orbital dynamics of stars in galactic potentials) and the numerical computing skills needed to run the calculations. The paper was published in *The Astronomical Journal* in 1964, when Heiles was still a graduate student — a remarkable debut by any measure.

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3. Why the Problem Matters

The Hénon–Heiles paper is important for reasons that extend well beyond the original astronomical question. It was one of the first clear numerical demonstrations that a deterministic Hamiltonian system — one with no friction, no randomness, and no external forcing — could produce motion that was, for all practical purposes, unpredictable. This was deeply counterintuitive. Classical mechanics had long been associated with clockwork regularity; Laplace had famously argued that a sufficiently powerful intellect, knowing all positions and velocities at one instant, could predict the entire future of the universe. Hénon and Heiles showed that even the simplest nonlinear system could exhaust Laplace's demon's supply of decimal places.

Their work fed directly into what became known as chaos theory in the 1970s and 1980s. It provided one of the clearest early numerical demonstrations of the *KAM theorem* (Kolmogorov–Arnold–Moser) in action — a deep result from pure mathematics that guarantees the survival of quasi-periodic, ordered motion for *most* initial conditions at low energies in nonlinear systems, while also predicting its breakdown as energy increases. The Hénon–Heiles system remains a standard benchmark for new numerical methods in Hamiltonian mechanics precisely because its behavior is so richly varied across its parameter space: ordered and integrable at low energy, mixed at intermediate energy, and largely chaotic at high energy, all within the same compact set of equations.

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4. The Mathematics

The Hamiltonian

The Hénon–Heiles system is described by a Hamiltonian — a function H representing the total energy of the system from which all equations of motion are derived. The system has two degrees of freedom: generalized coordinates q_1 and q_2 (two directions of displacement in a 2D potential well), and their conjugate momenta p_1 and p_2 (which here equal the corresponding velocities, since the mass is set to 1). The Hamiltonian is:

$$H(q_1, q_2, p_1, p_2) = \underbrace{\frac{p_1^2 + p_2^2}{2}}_{\text{kinetic}} + \underbrace{\frac{q_1^2 + q_2^2}{2}}_{\text{harmonic}} + \underbrace{\lambda\left(q_1^2 q_2 - \frac{q_2^3}{3}\right)}_{\text{cubic coupling}}$$

with $\lambda = 1$ (the canonical convention). The first two terms alone would describe two uncoupled simple harmonic oscillators — a completely integrable system with forever-periodic motion. The cubic term couples the two oscillators and is responsible for all the interesting nonlinear behavior. It has the symmetry of an equilateral triangle (invariant under 120° rotation), which is why the accessible region of configuration space at any given energy has a characteristic triangular shape — visible directly in Plot 2.

Equations of Motion

Hamilton's equations are derived by taking partial derivatives of H. The position equations are:

$$\dot{q}_1 = \frac{\partial H}{\partial p_1} = p_1 \qquad \dot{q}_2 = \frac{\partial H}{\partial p_2} = p_2$$

Since $\dot{q}_i = p_i$, the momenta are simply the velocities. The momentum (force) equations are:

$$\dot{p}_1 = -\frac{\partial H}{\partial q_1} = -q_1 - 2\lambda q_1 q_2$$

$$\dot{p}_2 = -\frac{\partial H}{\partial q_2} = -q_2 - \lambda(q_1^2 - q_2^2)$$

The linear terms $(-q_1, -q_2)$ are the harmonic restoring forces. The nonlinear terms couple the two coordinates: the displacement in q_1 affects the force in q_2 and vice versa. This coupling is what makes the system non-integrable in general. The four first-order ODEs together define a vector field in the 4D phase space (q_1, q_2, p_1, p_2), and the trajectory of the system is an integral curve of that field confined to the 3D energy surface $H = E$.

The Critical Energy Threshold

The cubic potential is unbounded: at large displacements the particle can escape to infinity. The potential energy surface has three equivalent saddle points, all occurring at energy $E = 1/6 \approx 0.1667$. Below this threshold the particle is permanently confined; above it, escape is possible. The qualitative character of the motion changes dramatically with energy:

Energy	Regime	Character
$E \ll 1/6$, e.g. $E = 1/12$	Subcritical	Quasi-periodic; smooth KAM tori; approximate third integral exists
$E \approx 1/8$	Mixed	Islands of regular motion surrounded by a chaotic sea
$E \geq 1/6$	Critical/supercritical	Predominantly chaotic; most tori destroyed; escape possible

This script uses $E = 1/12 \approx 0.0833$, which is comfortably below the threshold and places the system squarely in the ordered regime.

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5. How the Script Estimates the Solution

Why Standard ODE Solvers Fail Here

Hamiltonian systems possess a fundamental geometric property: they preserve the phase-space volume element $dq_1 \wedge dq_2 \wedge dp_1 \wedge dp_2$ at all times (Liouville's theorem), and every trajectory lies exactly on the 3D energy surface $H = \text{const}$ forever. A standard integrator such as fourth-order Runge–Kutta does not respect this structure. Over long integration times, RK4 slowly and systematically leaks or gains energy — the trajectory drifts off the true energy surface. For the Hénon–Heiles problem, which requires thousands of time units of integration to accumulate a useful Poincaré section, this drift would corrupt the results entirely. The Poincaré crossing points would scatter spuriously even for perfectly ordered trajectories, making it impossible to distinguish genuine chaos from numerical error.

The Yoshida 4th-Order Symplectic Integrator

The script uses the **Yoshida 4th-order symplectic integrator**, derived from a 1990 paper by Haruo Yoshida. A symplectic integrator is one that exactly preserves the symplectic 2-form $\omega = dq_1 \wedge dp_1 + dq_2 \wedge dp_2$ of the phase space at every time step. Geometrically, this means it is the discrete-time analog of a true Hamiltonian flow: it keeps the trajectory on a nearby *shadow* energy surface that deviates from the true Hamiltonian H by only $O(dt^4)$, and this deviation does not accumulate with time. Energy errors oscillate around a fixed small level rather than drifting — exactly what is seen in Plot 4.

The method is constructed by composing three weighted copies of the simple leapfrog (Störmer–Verlet) integrator. The Yoshida coefficients are derived from the requirement that error terms of order dt^2 and dt^3 cancel exactly, leaving a global error of order dt^4. Crucially, the coefficients depend only on $\sqrt[3]{2}$ and are **identical for any separable Hamiltonian** of the form $H = T(\mathbf{p}) + V(\mathbf{q})$ — which includes the Hénon–Heiles system, the pendulum, planetary orbits, and every other conservative mechanical system in this course. Each full time step has the structure:

$$\text{for } j = 0, 1, 2: \quad \mathbf{q} \leftarrow \mathbf{q} + c_j \mathbf{p}\, dt, \quad \mathbf{p} \leftarrow \mathbf{p} + d_j \mathbf{F}(\mathbf{q})\, dt$$

$$\text{then:} \quad \mathbf{q} \leftarrow \mathbf{q} + c_3 \mathbf{p}\, dt$$

where $\mathbf{q} = (q_1, q_2)$, $\mathbf{p} = (p_1, p_2)$, and $\mathbf{F}$ is the force vector evaluated at the already-updated position. This alternating drift–kick–drift structure is what makes the method symplectic.

Simulation Parameters

The script integrates **six trajectories** simultaneously, all at energy $E = 1/12$. All six share the initial conditions $q_2^0 = 0$ and $p_2^0 = 0$, with p_1^0 distributed across the accessible range from approximately 0.020 to 0.376, and q_1^0 computed from energy conservation as $q_1^0 = \sqrt{2E - (p_1^0)^2}$. Each trajectory runs for $t_{\text{final}} = 5000\,\text{s}$ with step size $dt = 0.01$, producing **500,000 integration steps** per trajectory. The Poincaré section accumulates approximately **750 crossings** per trajectory over this run.

6. Why These Four Plots

The four plots are not arbitrary diagnostics. Each illuminates a different aspect of the same underlying physics, and together they tell a complete story: how the system behaves in time, what region of physical space it explores, what its long-term geometric structure in phase space looks like, and whether the numerical method can be trusted to the degree necessary to believe any of the above.

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7. The Plots in Detail

Plot 1 — Time Series: $q_1(t)$ and $p_1(t)$ *(Cell 04 in henon_heiles.ipynb)*

This plot shows two of the four phase-space coordinates — the generalized coordinate q_1 (crimson) and its conjugate momentum $p_1 = \dot{q}_1$ (green) — as explicit functions of time, over the first 200 seconds of the simulation. The trajectory shown is the fourth of the six (index 3), which starts at $q_1^0 = 0.335$, $p_1^0 = 0.234$, $q_2^0 = 0$, $p_2^0 = 0$. To keep the curves smooth without overplotting, the 20,000 integration steps in this window are subsampled at stride 5, yielding 4,000 plotted points.

The oscillations are clearly not simple sinusoids: the amplitude of both q_1 and p_1 waxes and wanes on a slow timescale as energy sloshes back and forth between the q_1 and q_2 degrees of freedom — a phenomenon called *amplitude modulation* or beating. This slow envelope is the signature of two incommensurate frequencies: the fast oscillation within each well and the slow exchange between them. The ratio of these two frequencies is generally irrational in a nonlinear system, so the motion never exactly repeats. The fact that q_1 and p_1 are visibly correlated but not in phase is simply because $p_1 = \dot{q}_1$: the velocity leads the displacement by roughly a quarter cycle, just as in a simple harmonic oscillator, but the nonlinearity distorts this relationship.

Big-picture idea: Nonlinear quasi-periodic motion is neither simply periodic nor chaotic. It has recognizable structure — amplitude modulation, near-repetition, bounded oscillations — without ever exactly repeating. A student seeing this plot for the first time should understand that "structured complexity" is a third category of behavior distinct from both clockwork regularity and randomness.

Plot 2 — Configuration Space: q_1 vs. q_2 *(Cell 05 in henon_heiles.ipynb)*

This plot projects the four-dimensional phase-space trajectory onto the two-dimensional plane of physical positions (q_1, q_2) — the actual spatial coordinates, stripped of the momentum information. All six trajectories at $E = 1/12$ are overlaid in distinct colors. To keep the plot from becoming an opaque mass of lines, the 500,000-step trajectories are subsampled at stride 2000, giving 250 plotted points per trajectory (1,500 total line segments across all six).

Every trajectory is confined to the same bounded triangular region. This boundary is not a numerical artifact — it is the *zero-velocity curve* of the potential, the locus of points where all kinetic energy has been converted to potential energy: $V(q_1, q_2) = E$. A particle cannot penetrate beyond this curve because it would require negative kinetic energy. The triangular shape, with rounded corners and a flattened bottom, is a direct geometric consequence of the threefold symmetry of the cubic term $q_1^2 q_2 - q_2^3/3$ in the Hamiltonian. Note that the accessible region extends further along the q_2 axis than along q_1, reflecting the asymmetry of the cubic coupling — the q_2 saddle point is at a somewhat higher displacement than the two q_1 saddle points.

Big-picture idea: Energy conservation rigidly constrains which regions of physical space a trajectory may visit. The shape of the accessible region is determined entirely by the potential — by the mathematical form of the Hamiltonian — and is therefore the same for all six trajectories even though they start from different initial conditions. For a student accustomed to 1D oscillators, this plot makes viscerally clear that two-dimensional Hamiltonian systems have a rich spatial structure that one-dimensional systems simply cannot exhibit.

Plot 3 — Poincaré Section: q_2 vs. p_2 at $q_1 = 0,\ \dot{q}_1 > 0$ *(Cell 06 in henon_heiles.ipynb)*

The Poincaré section is the most information-dense of the four plots and the most directly connected to the mathematical theory of Hamiltonian chaos. Rather than plotting the trajectory continuously, we slice the four-dimensional phase space with a three-dimensional hyperplane — specifically $q_1 = 0$ — and record only the (q_2, p_2) coordinates of each crossing where $\dot{q}_1 = p_1 > 0$ (the trajectory is moving in the positive q_1 direction). This reduces the continuous 4D trajectory to a discrete set of 2D points; after 5,000 seconds of integration each trajectory contributes approximately 750 such points. No subsampling is applied — every detected crossing is plotted.

At $E = 1/12$, the crossing points for each of the six trajectories lie on a smooth, closed curve rather than scattering freely across the section. These closed curves are the Poincaré slice of a KAM torus: a 2D surface embedded in the 4D phase space on which the trajectory is forever confined. The curves nest concentrically, and they do not intersect one another — a consequence of the uniqueness theorem for Hamilton's equations, which guarantees that two distinct trajectories at the same energy can never pass through the same point in phase space. The innermost curve (corresponding to Trajectory 4, the one closest to the center of the nest) belongs to the trajectory with the most nearly equal partition of energy between the two degrees of freedom.

Near the center of the nested curves sits a *fixed point* — a location that would be visited by a perfectly periodic orbit, one that returns to exactly the same (q_2, p_2) position after every single crossing. The outermost curves belong to trajectories that are the most "unequally loaded" in their energy partition between q_1 and q_2.

Big-picture idea: Quasi-periodic motion leaves a geometric fingerprint on the Poincaré section:

smooth, closed invariant curves. Chaotic motion would instead produce a cloud of scattered points filling an area ergodically. The visual contrast between these two outcomes is one of the most powerful diagnostic tools in Hamiltonian dynamics. A student who reruns this script at $E = 1/6$ will watch these smooth curves dissolve into a scattered cloud, providing a visceral, direct demonstration of the KAM-to-chaos transition — the central result of the Hénon–Heiles paper itself.

Plot 4 — Energy Drift: $|H(t) - H_0|$ *(Cell 07 in henon_heiles.ipynb)*

This semilogarithmic plot shows how much the numerically computed total energy $H(t)$ deviates from its initial value $H_0 = E = 1/12$ over the entire 5,000-second integration of trajectory 4. The vertical axis spans roughly from 10^{-13} to 10^{-10}; the horizontal axis covers all 5,000 seconds. To make the oscillating envelope visible without rendering all 500,000 points, the data is subsampled at stride 250, giving 2,000 plotted points. The deviation is computed as the absolute value $|H(t) - H_0|$, evaluated by substituting the numerically computed (q_1, q_2, p_1, p_2) values back into the Hamiltonian formula at each stored step.

The energy error does not grow with time. It oscillates rapidly at a nearly constant level, bounded above by approximately 10^{-10} — more than nine orders of magnitude smaller than the energy itself (≈ 0.0833). The rapid up-and-down spikes visible in the plot are not numerical instability; they reflect the integrator sampling a *shadow Hamiltonian* that oscillates closely around the true H. Occasional deep downward spikes, reaching as low as 10^{-13}, occur when the shadow Hamiltonian momentarily coincides nearly exactly with the true one. A non-symplectic integrator such as RK4 would instead show the energy error growing monotonically — either drifting upward (trajectory spiraling outward) or downward (spiraling inward) — eventually growing large enough to qualitatively change the trajectory's behavior.

Big-picture idea: A symplectic integrator provides a qualitatively different kind of error guarantee than a general-purpose ODE solver. The question is not merely whether the instantaneous error is small, but whether it is *bounded for all time*. Only a symplectic method provides this guarantee, because only a symplectic method exactly preserves the geometric structure that Hamiltonian mechanics demands. This is the reason symplectic integrators are the gold standard for long-time simulation of conservative systems — from celestial mechanics to molecular dynamics to the Hamiltonian quantum simulation methods for which this course ultimately prepares you.

References

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